



F1-15-MKII Bernoulli's Theorem Demonstration

This accessory illustrates those circumstances to which Bernoulli's Theorem may be applied. Also, separately, why in other circumstances the theorem gives an inadequate description of the fluid behaviour.

PRODUCT CODE: F1-15-MKII

Categories: Agricultural, F1-15-MKII Bernoulli's Theorem Demonstration

Tags: Armfield , Bernoullis Theorem , Calibrating A Bourdon Type Pressure Gauge , Coefficient Of Discharge , Converging Duct , Diverging Duct , F1-15 , F1-15-MKII , Flow Measurement , Fluid Behaviour , Fluid Mechanics , Hydraulics Bench , Manometer , Pressurisation , Static Pressure , Venturi

DESCRIPTION

This accessory illustrates those circumstances to which Bernoulli's Theorem may be applied. Also, separately, why in other circumstances the theorem gives an inadequate description of the fluid behavior.

The test section consists of a classical Venturi machined in clear acrylic.

A series of wall tapplings enable measurement of the static pressure distribution along the converging and diverging duct. A total head tube is provided to traverse along the centre line of the test section. These tapplings are connected to a manometer bank incorporating a manifold with an air bleed valve.

Pressurisation of the manometers is facilitated by a hand pump. The test section is arranged so that the characteristics of flow through both a converging and diverging section can be studied. Water is fed through a hose connector and is controlled by a flow regulator valve at the outlet of the test section.

The Venturi can be demonstrated as a means of flow measurement and the discharge coefficient can be determined.

TECHNICAL SPECIFICATIONS

Manometer range: 0-400mm

Number of manometer tubes: 11

Throat diameter: 14mm

Upstream diameter: 26.0mm

Upstream taper: 21°
Downstream taper: 9°

Requires Hydraulics Bench Service unit F1-10/FI-10-2

FEATURES & BENEFITS

This unit has been updated and now:

- The classical venturi profile has been enhanced to industry standard, leading to better accuracy and efficiency in demonstrating Bernoulli's theorem. This improvement will affect the flow dynamics within the unit.
- More pressure tapings have been added, resulting in increased pressure drop resolution. This enhancement allows for a more detailed analysis of pressure variations within the system, offering a more comprehensive understanding of fluid dynamics.
- The manometer height has been increased. This modification suggests that the unit can now handle greater flowrates, potentially up to 20 L/min. This is crucial for experiments or demonstrations involving higher flow rates, providing versatility in the application of the unit.

These improvements collectively contribute to a more advanced and capable apparatus for demonstrating Bernoulli's theorem, offering better insights into fluid dynamics and providing a more comprehensive educational experience.

Experimental Content

- To investigate the validity of the Bernoulli equation when applied to the steady flow of water in a converging or a diverging duct
- Conservation of energy convergent/divergent pipe flow
- Effect of friction loss on Bernoulli equation
- Recording the pressure curve in a Venturi nozzle
- Recording the velocity curve in a Venturi nozzle
- Determining the flow coefficient

F1-15-MKII Training Video (English & Spanish)

English Language:

Vídeo en español:



Dimensions

Length: 0.80m

Width: 0.40m

Height: 0.65m

Order Codes

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